

Week 12 - Equivalence relations and groups

Definition: Let H be a subgroup of a group, G , and for $g \in G$ define $gH = \{gh \in G : h \in H\} \subset G$. Let G/H be the set of all possible subsets of the form gH ,

$$G/H = \{gH : g \in G\}.$$

2) Prove that $g \in H$ if and only if $gH = H$.

2) Prove that

$$G/H = G/\sim$$

for some equivalence relation $\sim$ on G .

3) Let $G = (\mathbb{R}^2, +, (0, 0)) =: \mathbb{R}^2$ and let $H = \{(x, 3x) \in \mathbb{R}^2 : x \in \mathbb{R}\}$.

(a) Prove that $H \subset G$ is a subgroup, and describe the set G/H . Include a sketch as well. (Since the operation is addition, for $v \in \mathbb{R}^2$, use the notation $v + H$ instead of vH .)

(b) Prove that the triple $(\mathbb{R}^2/H, [+], H)$ is a group, where for $v, w \in \mathbb{R}^2$,

$$(v + H)[+](w + H) = (v + w) + H.$$

4) If G is any group, and H is an arbitrary subgroup, is there always a well-defined operation, $m : G/H \times G/H \rightarrow G/H$, given by $m(gH, kH) = (gk)H$ that makes G/H into a group? Either prove yes, or explain why you think no.

5) Given a group G and $g \in G$, prove that there is a bijection $H \rightarrow gH$.

6) Suppose G is a finite group with n elements and H is a subgroup with k elements. Prove that the set G/H has n/k elements.

7) We say a subgroup $N \subset G$ is a **normal subgroup** if for any $g \in G$ and $n \in N$ we have $gng^{-1} \in N$. Prove that if N is a normal subgroup, then G/N is a group.

8) Prove that if $f : G \rightarrow G'$ is a group homomorphism, then the kernel of f , $\ker f = \{g \in G : f(g) = e'\}$, is a normal subgroup of G .

9) For a group homomorphism $f : G \rightarrow G'$, identify $G/\ker f$ as a set of the form $G/\sim$ for some equivalence relation $\sim$ that you have seen before in class.